



NOTE ON THE CARLEMAN'S INEQUALITY FOR A NEGATIVE POWER NUMBER

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ABSTRACT. By the method of indeterminate coefficients we prove the inequality

$$\sum_{n=1}^{\infty} \frac{n}{\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_n}} \leq 2 \sum_{n=1}^{\infty} \left(1 - \frac{1}{3n+1} - 4n^2 \sum_{k=n}^{\infty} \frac{1}{k(k+1)^2(3k+1)(3k+4)} \right) a_n,$$

where $a_n > 0$, $n = 1, 2, \dots$, $\sum_{n=1}^{\infty} a_n < \infty$.

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1. INTRODUCTION

The following Carleman inequality is well known (see [1, Chapter 9.12]).

$$(1.1) \quad \sum_{n=1}^{\infty} \left(\frac{a_1^{\frac{1}{p}} + a_2^{\frac{1}{p}} + \dots + a_n^{\frac{1}{p}}}{n} \right)^p \leq \left(\frac{p}{p-1} \right) \sum_{n=1}^{\infty} a_n,$$

where $a_n \geq 0$, $n = 1, 2, \dots$, $\sum_{n=1}^{\infty} a_n < \infty$, and $p > 1$.

Letting $p \rightarrow +\infty$, it follows from (1.1) that

$$(1.2) \quad \sum_{n=1}^{\infty} (a_1 a_2 \cdots a_n)^{\frac{1}{n}} \leq e \sum_{n=1}^{\infty} a_n.$$

In practice, the inequality (1.2) is strict; i.e.,

$$(1.3) \quad \sum_{n=1}^{\infty} (a_1 a_2 \cdots a_n)^{\frac{1}{n}} < e \sum_{n=1}^{\infty} a_n,$$

if $a_n \geq 0, n = 1, 2, \dots, 0 < \sum_{n=1}^{\infty} a_n < \infty$.

The constant e is sharp in the sense that it cannot be replaced by a smaller one.

Recently, the inequality (1.3) has also been improved by many authors, for example: Yang Bicheng and L. Debnath [2] with

$$(1.4) \quad \sum_{n=1}^{\infty} (a_1 a_2 \cdots a_n)^{\frac{1}{n}} < e \sum_{n=1}^{\infty} \left(1 - \frac{1}{2n+2}\right) a_n,$$

in [3] by Yan Ping and Sun Guozheng with

$$(1.5) \quad \sum_{n=1}^{\infty} (a_1 a_2 \cdots a_n)^{\frac{1}{n}} < e \sum_{n=1}^{\infty} \left(1 + \frac{1}{n+1/5}\right)^{-1/2} a_n,$$

and in [4] by X. Yang with

$$(1.6) \quad \sum_{n=1}^{\infty} (a_1 a_2 \cdots a_n)^{\frac{1}{n}} < e \sum_{n=1}^{\infty} \left(1 - \frac{1}{2(n+1)} - \frac{1}{24(n+1)^2} - \frac{1}{48(n+1)^3}\right) a_n,$$

and

$$(1.7) \quad \sum_{n=1}^{\infty} (a_1 a_2 \cdots a_n)^{\frac{1}{n}} < e \sum_{n=1}^{\infty} \left(1 - \sum_{k=1}^6 \frac{b_k}{(n+1)^k}\right) a_n,$$

where

$$b_1 = \frac{1}{2}, b_2 = \frac{1}{24}, b_3 = \frac{1}{48}, b_4 = \frac{73}{5760}, b_5 = \frac{11}{1280}, b_6 = \frac{1945}{580608},$$

$$a_n \geq 0, n = 1, 2, \dots, 0 < \sum_{n=1}^{\infty} a_n < \infty.$$

We rewrite the inequality (1.1) with $r = \frac{1}{p}$ as follows

$$(1.8) \quad \sum_{n=1}^{\infty} \left(\frac{a_1^r + a_2^r + \cdots + a_n^r}{n} \right)^{1/r} \leq (1-r)^{-1/r} \sum_{n=1}^{\infty} a_n,$$

where $a_n \geq 0, n = 1, 2, \dots, \sum_{n=1}^{\infty} a_n < \infty$ and $0 < r < 1$.

In [5], we have improved Carleman's inequality (1.8) for a negative power number $r < 0$ as follows

$$(1.9) \quad \sum_{n=1}^{\infty} \left(\frac{a_1^r + a_2^r + \cdots + a_n^r}{n} \right)^{1/r} \leq \begin{cases} (1-r)^{-1/r} \sum_{n=1}^{\infty} a_n, & \text{if } -1 \leq r < 1, r \neq 0, \\ \frac{r}{r-1} 2^{(r-1)/r} \sum_{n=1}^{\infty} a_n, & \text{if } r < -1, \end{cases}$$

where $a_n > 0, n = 1, 2, \dots, \sum_{n=1}^{\infty} a_n < \infty$.

In the case of $r = -1$, we obtain from (1.9) the inequality

$$(1.10) \quad \sum_{n=1}^{\infty} \frac{n}{\frac{1}{a_1} + \frac{1}{a_2} + \cdots + \frac{1}{a_n}} \leq 2 \sum_{n=1}^{\infty} a_n,$$

where $a_n > 0, n = 1, 2, \dots, \sum_{n=1}^{\infty} a_n < \infty$.

2. MAIN RESULT

In this paper, we shall prove the following theorem.

Theorem 2.1. Let $a_n > 0$, $n = 1, 2, \dots$, and $\sum_{n=1}^{\infty} a_n < \infty$. Then we have

$$(2.1) \quad \sum_{n=1}^{\infty} \frac{n}{\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_n}} \leq 2 \sum_{n=1}^{\infty} \left(1 - \frac{1}{3n+1} - 4n^2 \sum_{k=n}^{\infty} \frac{1}{k(k+1)^2(3k+1)(3k+4)} \right) a_n.$$

Remark 2.2. From the inequality (2.1), we obtain the following inequalities:

$$(2.2) \quad \sum_{n=1}^{\infty} \frac{n}{\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_n}} < 2 \sum_{n=1}^{\infty} \left(1 - \frac{1}{3n+1} - \frac{4n}{(n+1)^2(3n+1)(3n+4)} \right) a_n,$$

$$(2.3) \quad \sum_{n=1}^{\infty} \frac{n}{\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_n}} < 2 \sum_{n=1}^{\infty} \left(1 - \frac{1}{3n+1} \right) a_n,$$

and

$$(2.4) \quad \sum_{n=1}^{\infty} \frac{n}{\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_n}} < 2 \sum_{n=1}^{\infty} a_n.$$

Indeed, we note that the inequalities (2.2), (2.3), (2.4) are implied from (2.1), because

$$(2.5) \quad \begin{aligned} 1 - \frac{1}{3n+1} - 4n^2 \sum_{k=n}^{\infty} \frac{1}{k(k+1)^2(3k+1)(3k+4)} \\ < 1 - \frac{1}{3n+1} - \frac{4n}{(n+1)^2(3n+1)(3n+4)} \\ < 1 - \frac{1}{3n+1} < 1. \end{aligned}$$

Hence, we obtain from (2.1), (2.5) that

$$(2.6) \quad \begin{aligned} \sum_{n=1}^{\infty} \frac{n}{\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_n}} \\ \leq 2 \sum_{n=1}^{\infty} \left(1 - \frac{1}{3n+1} - 4n^2 \sum_{k=n}^{\infty} \frac{1}{k(k+1)^2(3k+1)(3k+4)} \right) a_n \\ = 2 \sum_{n=1}^{\infty} \left(1 - \frac{1}{3n+1} - \frac{4n}{(n+1)^2(3n+1)(3n+4)} \right. \\ \left. - 4n^2 \sum_{k=n+1}^{\infty} \frac{1}{k(k+1)^2(3k+1)(3k+4)} \right) a_n \end{aligned}$$

$$\begin{aligned}
&= 2 \sum_{n=1}^{\infty} \left(1 - \frac{1}{3n+1} - \frac{4n}{(n+1)^2(3n+1)(3n+4)} \right) a_n \\
&\quad - 8 \sum_{n=1}^{\infty} \left(\sum_{k=n+1}^{\infty} \frac{1}{k(k+1)^2(3k+1)(3k+4)} \right) n^2 a_n \\
&< 2 \sum_{n=1}^{\infty} \left(1 - \frac{1}{3n+1} - \frac{4n}{(n+1)^2(3n+1)(3n+4)} \right) a_n \\
&< 2 \sum_{n=1}^{\infty} \left(1 - \frac{1}{3n+1} \right) a_n \\
&< 2 \sum_{n=1}^{\infty} a_n.
\end{aligned}$$

To prove Theorem 2.1, we first prove the following lemma.

Lemma 2.3. *We have*

$$(2.7) \quad \frac{1}{\frac{1}{a_1} + \frac{1}{a_2} + \cdots + \frac{1}{a_n}} \leq \frac{a_1 b_1^2 + a_2 b_2^2 + \cdots + a_n b_n^2}{(b_1 + b_2 + \cdots + b_n)^2},$$

where $a_k > 0$, $b_k > 0$, $\forall k = 1, 2, \dots, n$.

Proof. This is a simple application of the Cauchy-Schwartz inequality

$$\begin{aligned}
(2.8) \quad (b_1 + b_2 + \cdots + b_n)^2 &\leq \left(\frac{1}{\sqrt{a_1}} \sqrt{a_1} b_1 + \frac{1}{\sqrt{a_2}} \sqrt{a_2} b_2 + \cdots + \frac{1}{\sqrt{a_n}} \sqrt{a_n} b_n \right)^2 \\
&\leq \left(\frac{1}{a_1} + \frac{1}{a_2} + \cdots + \frac{1}{a_n} \right) (a_1 b_1^2 + a_2 b_2^2 + \cdots + a_n b_n^2).
\end{aligned}$$

□

Proof of Theorem 2.1. We prove the theorem by the method of indeterminate coefficients.

Consider $b_1, b_2, \dots$ to be the positive indeterminate coefficients. Let $N = 1, 2, \dots$. Put

$$(2.9) \quad C_k = \sum_{n=k}^N \frac{n b_k^2}{(b_1 + b_2 + \cdots + b_n)^2}, \quad 1 \leq k \leq N.$$

Applying Lemma 2.3, we obtain

$$\begin{aligned}
(2.10) \quad \sum_{n=1}^N \frac{n}{\frac{1}{a_1} + \frac{1}{a_2} + \cdots + \frac{1}{a_n}} &\leq \sum_{n=1}^N \sum_{k=1}^n \frac{n a_k b_k^2}{(b_1 + b_2 + \cdots + b_n)^2} \\
&= \sum_{k=1}^N \sum_{n=k}^N \frac{n a_k b_k^2}{(b_1 + b_2 + \cdots + b_n)^2} \\
&= \sum_{k=1}^N C_k a_k.
\end{aligned}$$

Choosing $b_k = k$, $k = 1, 2, \dots$, we have from (2.9) that

$$(2.11) \quad C_k = \sum_{n=k}^N \frac{n k^2}{(1 + 2 + \cdots + n)^2} = 4k^2 \sum_{n=k}^N \frac{1}{n(n+1)^2}.$$

On the other hand, we have the equality

$$(2.12) \quad \frac{1}{2n^2 + \frac{2}{3}n} - \frac{1}{2(n+1)^2 + \frac{2}{3}(n+1)} - \frac{1}{n(n+1)^2} \\ = \frac{2}{n(n+1)^2(3n+1)(3n+4)} > 0, \quad \text{for all } n = 1, 2, \dots$$

Hence, it follows from (2.11) that

$$(2.13) \quad \sum_{n=k}^N \frac{1}{n(n+1)^2} \\ = \frac{1}{2k^2 + \frac{2}{3}k} - \frac{1}{2(N+1)^2 + \frac{2}{3}(N+1)} - \sum_{n=k}^N \frac{2}{n(n+1)^2(3n+1)(3n+4)} \\ \leq \frac{1}{2k^2 + \frac{2}{3}k} - \sum_{n=k}^N \frac{2}{n(n+1)^2(3n+1)(3n+4)}, \quad 1 \leq k \leq N.$$

Hence, we obtain from (2.10), (2.13) that

$$(2.14) \quad C_k = 4k^2 \sum_{n=k}^N \frac{1}{n(n+1)^2} \leq 2 - \frac{2}{3k+1} - 8k^2 \sum_{n=k}^N \frac{1}{n(n+1)^2(3n+1)(3n+4)}.$$

$$(2.15) \quad \sum_{n=1}^N \frac{n}{\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_n}} \\ \leq \sum_{k=1}^N C_k a_k \\ \leq \sum_{k=1}^N \left(2 - \frac{2}{3k+1} - 8k^2 \sum_{n=k}^N \frac{1}{n(n+1)^2(3n+1)(3n+4)} \right) a_k \\ \leq \sum_{k=1}^N \left(2 - \frac{2}{3k+1} \right) a_k - 8 \sum_{k=1}^N \left(\sum_{n=k}^N \frac{1}{n(n+1)^2(3n+1)(3n+4)} \right) k^2 a_k \\ = \sum_{k=1}^N \left(2 - \frac{2}{3k+1} \right) a_k - 8 \sum_{n=1}^N \left(\sum_{k=1}^n \frac{k^2 a_k}{n(n+1)^2(3n+1)(3n+4)} \right) \\ = \sum_{k=1}^N \left(2 - \frac{2}{3k+1} \right) a_k - 8 \sum_{n=1}^N \beta_n.$$

where

$$(2.16) \quad \beta_n = \frac{\sum_{k=1}^n k^2 a_k}{n(n+1)^2(3n+1)(3n+4)}.$$

We have

$$\begin{aligned}
 (2.17) \quad 0 &< \beta_n \\
 &= \frac{\sum_{k=1}^n k^2 a_k}{n(n+1)^2(3n+1)(3n+4)} \\
 &\leq \frac{n^2 \sum_{k=1}^n a_k}{9n^5} \\
 &= \frac{1}{9n^3} \sum_{k=1}^n a_k \sim \frac{1}{9n^3} \sum_{k=1}^{\infty} a_k, \text{ as } n \rightarrow +\infty.
 \end{aligned}$$

Hence, the series $\sum_{n=1}^{\infty} \beta_n$ converges. Letting $N \rightarrow +\infty$ in (2.15), we obtain

$$\begin{aligned}
 (2.18) \quad &\sum_{n=1}^{\infty} \frac{n}{\frac{1}{a_1} + \frac{1}{a_2} + \cdots + \frac{1}{a_n}} \\
 &\leq \sum_{k=1}^{\infty} \left(2 - \frac{2}{3k+1}\right) a_k - 8 \sum_{n=1}^{\infty} \beta_n \\
 &= \sum_{k=1}^{\infty} \left(2 - \frac{2}{3k+1}\right) a_k - 8 \sum_{n=1}^{\infty} \left(\sum_{k=1}^n \frac{k^2 a_k}{n(n+1)^2(3n+1)(3n+4)} \right) \\
 &= \sum_{k=1}^{\infty} \left(2 - \frac{2}{3k+1}\right) a_k - 8 \sum_{k=1}^{\infty} \left(\sum_{n=k}^{\infty} \frac{1}{n(n+1)^2(3n+1)(3n+4)} \right) k^2 a_k \\
 &= 2 \sum_{k=1}^{\infty} \left(1 - \frac{1}{3k+1} - 4k^2 \sum_{n=k}^{\infty} \frac{1}{n(n+1)^2(3n+1)(3n+4)}\right) a_k.
 \end{aligned}$$

The proof of Theorem 2.1 is complete. $\square$

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